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Recent progress of 3D printed vascularized tissues and organs

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ABSTRACT

Since the need for vascular networks to supply oxygen and nutrients while expelling metabolic waste, most cells can only survive within 200 μm of blood vessels; thus, the construction of well-developed blood vessel networks is essential for the manufacture of artificial tissues and organs. Three-dimensional (denoted as 3D) printing is a scalable, reproducible and high-precision manufacturing technology. In the past several years, there have been many breakthroughs in building various vascularized tissues, greatly promoting the development of biological tissue engineering. This paper highlights the latest progress of 3D printed vascularized tissues and organs, including the heart, liver, lung, kidney, and penis. We also discuss the application status and potential of the above printed tissues, and prospect the further requirement of 3D printing technology for manufacturing clinically useable vascularized tissues.

1. Introduction

The current transplanted tissues and organs for the repair of organ defects are mainly obtained from human donors [1]. Since the capabilities for regeneration of tissues and organs are strictly limited, it is significantly challenging to keep up with the clinical demands [2]. In China, there were approximately 1.5 million patients waiting for organ transplantation each year, but less than 1500 of them had access to suitable donors [3]. Additionally, the supply of tissues and organs has been nearly stagnant for a long time, while the demand for transplants continues to increase [4]. Therefore, the shortage of tissues and organs has become a major medical problem and a huge social burden.

In 1993, Prof. Vacanti came up with tissue engineering, which aims to create bionic tissues and organs to completely solve the problem of lack of donors [5]. Since the birth of the tissue engineering concept, numerous studies have been conducted in this field, however, only a few of achievements have been used in clinical transplantation, which mainly originates from limited technology in the construction of vascularized tissue [3]. Generally speaking, the distance of capillaries within living tissue is approximately 60–300 μm, and cells survive within a range of 200 μm surrounded by capillaries [6]. An area beyond this scope would lead to a dramatic decline of cell survival rates, because the vascular network is critical to oxygen diffusion along with nutrient supplementation and waste disposal. Thus, well-developed intravascular networks are pivotal for tissues and organs to survive long-term [7].

The past few years have witnessed some advances in tissue engineering technologies for creating vascular networks, such as microfluidics, organ-on-chip, and 3D printing [8–10]. Microfluidics technologies can accurately characterize the cellular survival microenvironment in all aspects, such as fluid flow profile, growth factor gradient, biomaterial mechanical property, which is facilitate to fabricate vascularized micro-tissues [8]. Organ-on-chip is a microfluidic chip bionic system based on micro-processing technology, which simulates the complex microstructure, microenvironment, and physiological function of specific human organs. The microchannel size of organ-on-chip can be comparable to the size of living cell, and the composition and temperature of the microenvironment can be accurately controlled, to simulate the situation of the extracellular matrix as much as possible, and enhance

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the reliability and operability of the experiment [9]. Moreover, it can design different two-dimensional or 3D structures and precision machining into microelectrodes and other devices to achieve cell cultivation, positioning, sequencing, mapping, and detection. It can further realize the unit operation of concentration gradient dilution, automatic sampling, cell culture, cell stimulation and labeling, and cell morphology and function detection in one body, and realize the high-connotation screening of cell multiple parameters, breaking the limitations of cell culture and model animal experiments. However, the two above technologies are mostly used for the vascularized tissue model construction in vitro [10]. 3D printing could fabricate artificial tissues and organs with pre-designed microstructure from 3D digital models, showing great potential for synchronously constructing well-developed vascular networks [11–13]. Typically, the first step is obtaining images through various imaging techniques, and then converting the image into 2D patterns by computer-aided technology for printing through the layer-by-layer process, which leads to precise deposition of biomaterials and even cells to the targeted location [14,15]. With the development of digital control technology, the current printed tissues already have good controllability and reproducibility, which is essential for industrialized production [16–18]. Therefore, the 3D printing, which allows for fully mimicking the human anatomic microstructure in theory, is the most promising technology for manufacturing tissues and organs with elaborate vascular networks [19,20].

Although artificial organs that can be used for clinical transplantation are still not available and more effort needs to be made, the past several years have witnessed the considerable development of 3D printing, and many 3D printed tissues and organs have been emerged [21–26], offering useful biological models for testing drug toxicity, efficacy, and chemotherapy [6,27–31]. Given the ethical concerns and the expense of animal testing, the replacement of corresponding animals by 3D printed models could not only reduce the number of experimental animals, but also keep down costs and increase the efficiency of drug discovery. 3D printing thus opens new avenues for the forward development of tissue engineering and provides hope for its clinical transformation.

Considering the rapid development and huge superiority of 3D printing for fabricating vascularized tissues and organs, this review focused on the most advanced achievements and application status of 3D printed vascularized models, including heart, liver, kidney, lung, and penis tissues. Moreover, the limitations and challenges for translating the printed vascularized tissues and organs from scientific research to clinical treatment were highlighted. We also briefly discuss the prospective solutions for accelerating developments of printed vascularized objects, and look forward to vascularized tissues and organs to meet human clinical needs.

2. 3D printed vascularized tissues and organs

The microvascular system plays an important role at the tissue and cellular levels, transporting oxygen and nutrients while also discharging metabolic waste [32,33]. To simulate the function of human tissues and organs, vascular networks in artificial tissues and organs should possess two following characteristics: high precision and hierarchical architecture [34]. Firstly, high precision is required because of the limited ability (typically less than 200 μm) of oxygen and nutrients to diffuse from capillaries to surrounding tissues. Low precision may cause the distance between capillaries and cells to be larger than the set value, which affects the cell's access to nutrients and oxygen, leading to apoptosis and tissue necrosis [35]. Secondly, as shown in Fig. 1, it is essential to construct hierarchical architectures containing arteries (diameter: normally more than 1 mm) and veins (diameter: normally more than 1 mm) as well as capillaries (diameter: normally between 10 and 15 μm). Additionally, it should be noted that the diameters of arterioles and venules, connected with large blood vessels and capillaries, typically range from 0.3 to 1 mm [3,36]. The arteries should be constructed to withstand high blood pressure and function as blood transporters to the targeted tissues, while the veins are fabricated to direct blood back to the heart. The capillaries are designed to provide nutrients and oxygen for most cells, while also connecting arteries and veins. Strictly precise geometrical and functional control is thus pivotal for fabricating vascularized tissues and organs, especially at the submicron level [17].

For constructing an excellent vascularized network, the types of cells that make up various blood vessels should be further considered (Fig. 1) [36,37]. Generally, capillaries are made up of endothelial cells (ECs) and

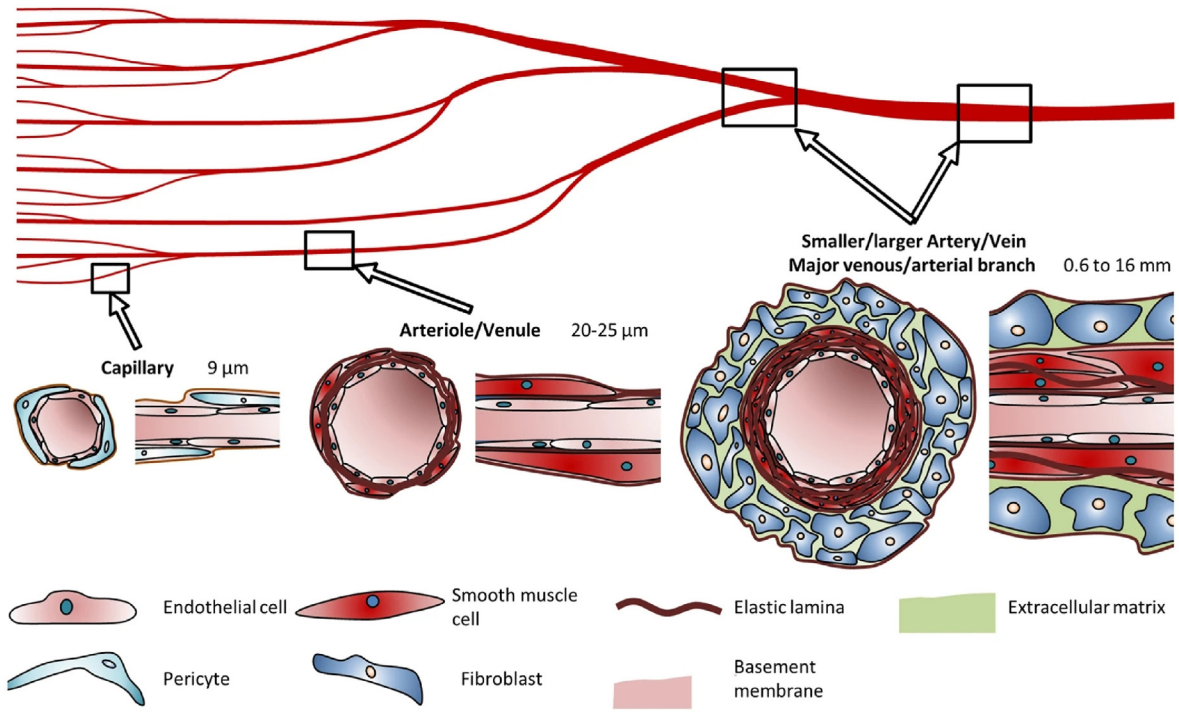


Fig. 1. Scale, cell type and composition of diverse blood vessels [36]. Adapted from an open-access source.

pericytes, while arteries and veins are composed of ECs, smooth muscle cells (SMCs), and pericytes. The main types of ECs currently used in the field of tissue engineering include human umbilical cord vein endothelial cells (HUVECs), human microvascular endothelial cells (HMVECs), and induced pluripotent stem cell derived endothelial cells (iPSC-ECs), which are the primary cell type that make up the inner lining of blood vessels. SMCs, pericytes, mesenchymal stem cells (MSCs), and fibroblasts are referred to as supporting cell types, as they are typically utilized for the stabilization of blood vessels in the field of tissue engineering.

Since the first extrusion 3D printer was created in 2003, 3D printing technology has flourished for 20 years and has become increasingly important in the field of biological tissue engineering (see Table 1) [3]. So far, 3D printing technology applied in the field of biological tissue engineering is mainly divided into three categories: droplet-based printing, extrusion-based printing, and laser-assisted printing [6]. At present, 3D printing technology faces two major challenges in the construction of vascularized tissues and organs, namely structural complexity, and component heterogeneity. Extrusion-based printing is a widely adopted strategy for vascularized tissues and organs construction, with the advantages of high cell density and large-sized structures. It takes extruded living cell-carrying biomaterial microfilaments as the unit, and the microfilaments are generally more than 100 μm in diameter. Therefore, the disadvantage of extrusion-based 3D printing is the low printing accuracy. In addition, extrusion-based printing is usually formed in layers on a plane, and it is still challenging to achieve structural features, such as cantilever beams and internal cavities. Droplet-based printing could make use of extremely small microdroplets, with the advantages of high cell viability and precision deposition, which is in favor of fine two-dimensional pattern-printing. However, such living cell-carrying droplets generally have low viscosity, and if they cannot be crosslinked in time during deposition, they are easily lost and significantly affect printing accuracy and structural complexity. In addition, the living cell density and structural sizes are generally limited. For the two types of material unit deposit-based technologies, droplet-based printing and extrusion-based printing, the main strategy to achieve the compositional heterogeneity of the print structure is the use of multiple ink nozzles or multi-ink control with multi-channel microfluidics during the printing process. Of course, multi-nozzle linkage printing is also a printing method derived from multi-material printing, which has shown great potential in terms of structural composition heterogeneity.

The significant advantages of laser-assisted printing are strong 3D forming ability and high precision. Classical stereolithography and digital light processing technology can achieve tens of micron printing accuracy of bio-hydrogel, and the two-photon printing method can even construct micron scale extracellular matrix structure [6,14]. However, light stimulation, as well as the necessary light absorbers and initiators in the liquid tank, can significantly affect the activity of the cells in the ink. Besides, the method usually only permits the addition of one ink component to the cylinder, greatly limiting the heterogeneous forming ability of this technology. In addition, the microstructures may become distorted or shrink if excessively exposed to laser energy. At present, in digital light processing, the crosslinking degree of different layers of inks was adjusted mainly by changing the gray level or light intensity of each mask pattern, resulting in a structure with heterogeneous crosslinking degree and mechanical properties in the Z-axis direction. Alternatively, the formation platform can switch between liquid cylinders containing various inks to achieve a switch between different materials or cells.

As previously stated, each of the current 3D printing technologies has limitations that prevent the selection of a single technology for universal use. In the future, a hybrid printer that combines the precision of laser-based spray printing with droplets or streams of cellular material could be a potential solution. If a support gel is used during the spray process to hold the cells, potential thermal issues can be reduced while ensuring precise material deposition for the final print.

Table 1
Examples of 3D printing techniques for fabricating tissues and organs.

tissues or organs	3D printing techniques	comments	ref
cardiac patches	extrusion-based 3D printing	the first to use fully personalized, non-supplemented bioinks	[43]
vascularized heart	FRESH	embedding printable hydrogels in a supporting bath	[44]
heart with various scales	FRESH v2.0	achieving the printing of 20 μm collagen filaments	[45]
full-size human heart	FRESH	FRESH can be used for fabricating large-scale organ	[48]
cardiac construct	extrusion-based 3D printing	the first to prepare human tropoelastin-based bioink	[39]
cardiac tissue	pneumatic-extrusion with a thermoplastic printhead	early vascular cell spheroids are favorable for fabricating vascular networks in myocardial tissues	[49]
heart	SPIRIT strategy	unparalleled bioprinting capability to replicate the complex organ geometry and internal structure in a faster manner	[22]
cardiac tissue	extrusion-based 3D printing	electrical stimulation is favorable for vascularization	[50]
liver tissues	four-arm branched vascular network 3D printing strategy	the obtained tissue models possess larger blood vessels and suitable cell density	[69]
liver tissues	extrusion-based 3D bioprinting	hepatocytes as bioink could be able to better simulate the microenvironment in vivo	[70]
liver tissue	extrusion-based 3D bioprinting	printed tissue showed vascularized capability after implantation	[71]
liver tissue	extrusion-based 3D bioprinting	biomaterial could fulfill the mechanical properties of 3D printing and the microenvironment required for living cells	[72]
liver tissue constructs	microfluidics-based single nozzle printhead and computer-controlled pneumatic pressure valves	solve alignment issues during switching multiple nozzles	[75]
kidney	extrusion-based 3D bioprinting	rapid and high-throughput generation of kidney organoids with highly reproducible cell number and viability	[89]
kidney tissue	extrusion-based 3D bioprinting	opens new avenues for fabricating vascularized tissues	[93]
alveolar model	3D printing strategy adopts a projector emitting light from below to display successive 2D slices of the model	the printed 3D structures are soft and biocompatible, and have a fine internal structure (resolution: 10–50 μm)	[107]
ultra-thin three-alveolar model	inkjet cell-printing	enable precise layer-by-layer regulation of spatial arrangement and population of various kinds of cells	[108]
triple-layered human	inkjet 3D printing with polyvinylpyrrolidone-	improve the stability and repeatability of the 3D printing process	[109]

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Table 1 (*continued*)

tissues or organs	3D printing techniques	comments	ref
alveolar lung model	based bioink and a 300 μm nozzle		
vascularized lung model	microextrusion-based 3D bioprinting	effectively vascularized lung cancer model	[110]
biodegradable scaffolds of penile corpus cavernosa	extrusion-based 3D printing	an important step towards the development of vascularized bioengineering scaffolds	[125]

2.1. Heart

In the human circulatory system, the heart is the power source that promotes blood flow. Blood vessels of the heart, including arteries, capillaries, and veins, are the pipeline system through which blood flows. The heart consists of four elastic chambers: the left atrium, the left ventricle, the right atrium, and the right ventricle. The heart muscle contracts to pump blood through the four chambers of the heart. The returning venous blood enters the right atrium and then flows back to the right ventricle through the superior and inferior vena cava. The right ventricle pumps blood into the pulmonary circulation to oxidize the blood to arterial blood through the pulmonary vein into the left atrium and into the left ventricle. The left ventricle is the strongest pump since it moves arterial blood through arterial vessels to all tissues. During ventricular contraction, the mitral and tricuspid valves close to prevent blood from flowing back into the heart chamber. When ventricular diastole occurs, the pulmonary and aortic valves close to prevent blood from flowing back into the ventricle. The heart has eight main blood vessels, including aorta, pulmonary artery, superior vena cava, inferior vena cava, pulmonary vein, cardiac vein, and the left and right coronary arteries. These vessels work together to maintain blood supply, ensure the heart's rhythmic contraction and relaxation, and promote blood circulation. Therefore, the vascular network plays an extremely important role in the heart. In view of the benefits of 3D printing technology in constructing intricate microstructures, 3D-printed vascularized heart tissues are gaining significant attention [38–42]. In 2019, Professor Dvir printed a living heart with a network of blood vessels, cells, and ventricles for the first time, garnering global interest [43]. In this research, a personalized hydrogel was delicately manufactured by using extracellular matrix produced during the differentiation of pluripotent stem cells (denoted as PSCs). The obtained cardiomyocytes and ECs, differentiated from PSCs, were then combined with the above personalized hydrogel to make bioinks for the printing of parenchymal cardiac tissue and blood vessels, respectively. They printed a cellularized heart which consisted of natural chambers and major vascular networks, and the size was like that of a cherry. Furthermore, the PSCs were derived from the patient's own adipose tissue, thus the vascularized cardiac tissue could fully match the immunological properties. However, for the time being, it could not pump blood, and the vascular networks cannot fully meet the demand for blood supply.

Collagen, a main ingredient of the anthropic extracellular matrix, has a natural immune compatibility advantage as bioinks for 3D printing. However, due to the unique solidification and soft properties of collagen, it has been challenging to construct functional vascularized tissues and organs. In 2019, Prof. Feinberg upgraded the freeform reversible embedding of suspend hydrogels (denoted as FRESH) strategy to FRESH v2.0 for the successful realization of 3D printing of collagen with high precision, opening new avenues for the fabrication of vascularized heart [44–46]. FRESH strategy was reported in 2015, the most important innovation is to embed printable hydrogels in a supporting bath, which was composed of special gelatin particles. The gelatin microparticles behave as a viscous fluid at high shear stresses, while behaving as a rigid body at low shear stresses [44]. Therefore, the needle-like nozzle is able to move smoothly during printing, while printed materials could be fixed in place once extruded from the nozzle, allowing successful printing of soft materials such as collagen. Different from gelation micro particles used by FRESH (approximately 250 μm in diameter), uniform spherical gelation micro particles (approximately 25 μm in diameter) were adopted as support bath in FRESH v2.0 [45]. Additionally, FRESH v2.0 adopted pH to control collagen gelation and improve printing resolution [47], achieving the printing of 20 μm collagen filaments. As shown in Fig. 2 (A-D), by using advanced FRESH v 2.0, they fabricated a porous microstructure for high-efficiency cellular infiltration and vascularization, and a heart model with patient-specific anatomical structure was successfully constructed. What's more, the results showed that FRESH v2.0 printed ventricles had many important properties, such as synchronous contraction and directed action potential propagation [45]. At present, FRESH v2.0 has realized 3D printing of a full-sized human heart model [48]. However, fabricating smaller capillaries (5–10 μm) using FRESH presents a huge challenge because the diameter of a normal nozzle exceeds 10 μm.

In 2020, Annabi et al. come from the University of California first prepared a human tropoelastin-based bioink using methacryloyl-substituted gelatin and tropoelastin [39]. The vascularized cardiac tissues printed by the above bioink showed endothelial barrier function, and cardiac muscle cells could beat spontaneously. Additionally, the tissues and exhibited excellent immunocompatibility in vivo, demonstrating the potential of the novel bioink for fabricating vascularized cardiac tissue. In 2022, Prof. Le and Prof. Liu et al. come from Tongji University first found that the early vascular cell spheroids are more favorable for fabricating vascular networks than single early vascular cells in 3D-printed myocardial tissues [49]. Using a fibrin-based bioink containing cardiomyocytes and early vascular cell spheroids, they printed a well-organized heart tissue with a high density of capillaries.

For better printing of elaborate vascular networks in artificial tissues and organs, in 2023, Prof. Xiong et al. proposed a sequential printing in a reversible ink templet (denoted as SPIRIT) strategy [22]. As shown in Fig. 2 (E), bio-inks were first printed into a suspension medium for the construction of external structures. Then, the sacrificial ink, with reversible self-healing properties, was printed into the first printed structure. Finally, the printed model was crosslinked in situ, while the suspended medium and sacrificial ink were removed to fabricate a free-form vascular network. Obviously, SPIRIT required a specific bioink that acts as both bioink and suspension medium. It was demonstrated that the SPIRIT technique was able to print ventricle models with perfusable vascular networks by using a microgel-based biphasic bioink containing human-induced PSCs. More importantly, the SPIRIT technique exhibited unparalleled print speed in constructing complex tissues, which is critical for large-scale applications. This brings 3D-printed heart tissues closer to meeting the demand of clinical transplantation. Additionally, the team also proposed achieving vascularized cardiac tissues by using electrical stimulation [50]. They found that the electrical stimulation not only significantly promoted elongation, migration, and interconnection of HUVECs in 3D printed heart tissue, but also enhanced the interaction between cardiomyocytes and HUVECs.

The advancement of 3D printed heart tissues is promising to revolutionize the pharmaceutical paradigm due to the ability of printed tissue to simulate the physiological environment [51–54]. In 2022, Hortigon-Vinagre et al. designed an organ-on-chip platform, which could evaluate the curative effects and side effects of newly developed drugs for treating cardiovascular diseases [55]. Recently, a lot of achievements have emerged in the field of 3D printed heart tissues [56–60]. However, printing a fully functional heart for transplantation has not been realized, and there is a long way for printed heart tissues to be used clinically. Firstly, it is necessary to improve the survival rates of cells and maintain the printed heart model's function, of which all the cells within printed tissues should be able to obtain oxygen and nutrients smoothly. Thus, fully biomimetic vascularization is essential, and current 3D printing techniques should be further improved for precisely printing capillaries,

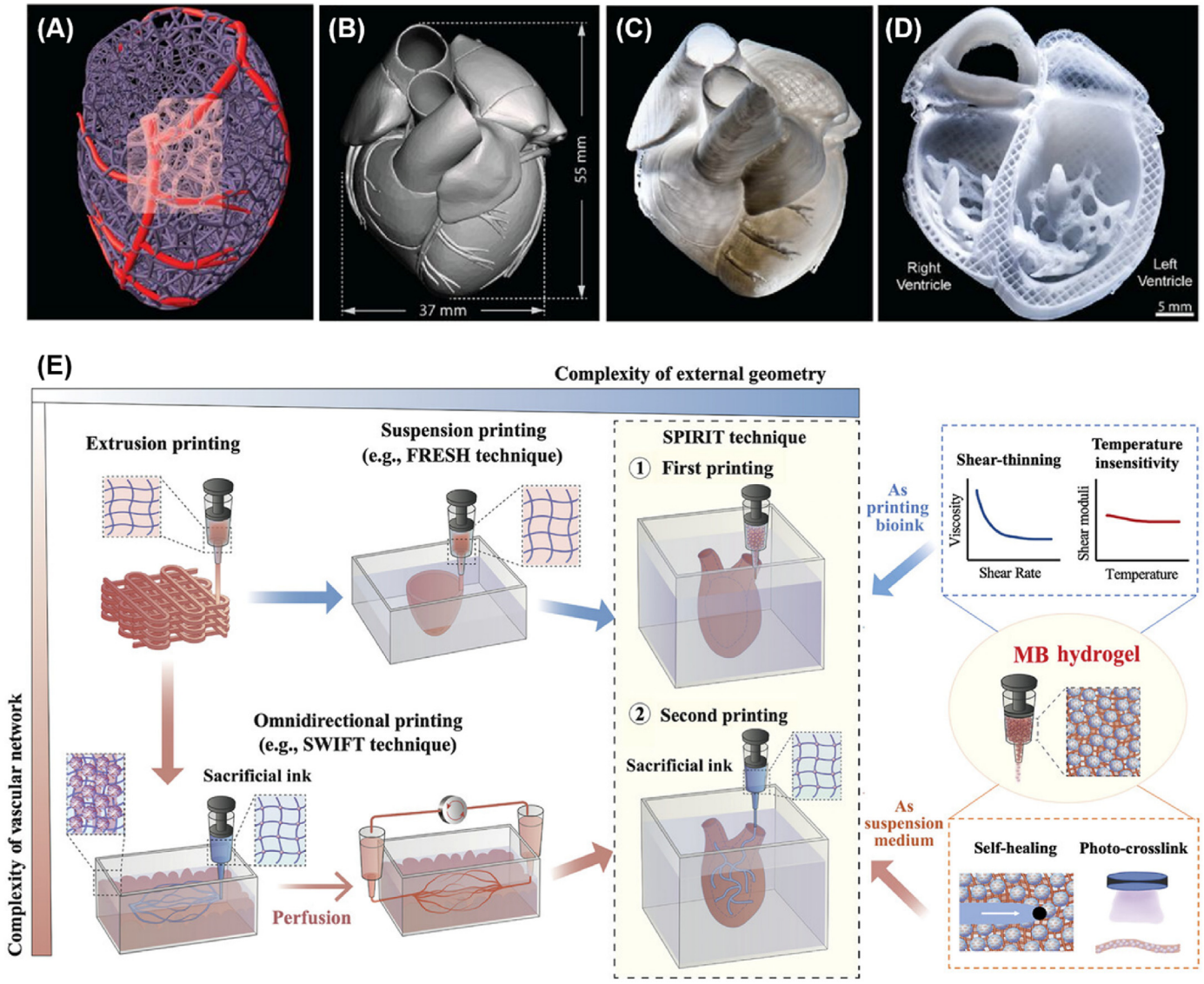


Fig. 2. (A–D) FRESH 3D printed vascularized heart model [45]. (A) The red, purple, and pink color represent Left ventricle, computationally fabricated vasculature, and subregion of interest, respectively. (B) Neonatal heart size. (C) FRESH-printed heart by using collagen-based ink. (D) Cross-sectional view of printed heart. (E) 3D printing of vascularized heart tissue with complex external geometries by SPIRIT strategy [22]. Reproduced with permission. Copyright 2019 [45], American Association for the Advancement of Science, and Copyright 2023 [22], John Wiley and Sons.

especially the smaller ones. Secondly, although most of the vascularized tissues were printed with biocompatible materials, biological rejection may still occur. Therefore, it is urgent to develop fully biocompatible bio-inks to completely avoid possible rejection and achieve physiological function. Looking back at the breakthroughs achieved in 3D printed heart tissues, we have reasons to imagine that the 3D printed fully functional heart is not far-fetched, of course, cooperation of scientists from various fields is very necessary.

2.2. Liver

The liver has a complex cross-scale vascular system, including the main vascular network formed by the hepatic vein, portal vein and hepatic artery and the microvascular network formed by hepatic blood sinuses. At the same time, the liver vascular system is very densely distributed, and the blood supply is rich, and 30 % of the total amount of living blood passes through the liver every minute. Among them, 75 % of the blood of the liver enters the liver from the portal vein, and the nutrients and toxins produced by the peripheral organs, including the intestine, spleen, pancreas, *etc*., are transported to the liver through the portal vein. This blood supplies nutrients to liver cells, and toxins in the blood are further processed by liver cells. The remaining 25 % of the blood is fed through the liver arteries, which branch off the abdominal arteries and provide oxygen to the liver. The peripheral blood flows from the portal vein and hepatic artery into the central vein through the narrow hepatic sinuses located at the intersection of the interlobular vein and interlobular artery. It then flows from the central vein into the hepatic vein and is finally reintroduced into the systemic circulation through the inferior vena cava [61–64]. Therefore, the fabrication of blood vessel networks is essential for manufacturing an artificial liver. In recent years, 3D printing has been the most distinguished representative of tissue engineering technology to fabricate artificial vascularized liver tissue. This technology may help resolve the problem of liver failure, one of the main causes of human mortality [65–67]. In 2016, Nguyen established a 3D printed human liver model by combining non-parenchymal cells with patient-derived hepatic cells [68]. It was used to test a kind of hepatotoxic drug, and the experimental results are in good agreement with the clinical data, demonstrating the effectiveness of 3D

printed liver models in simulating drug response. Although the 3D printed liver tissue mimetics could not fully mimic the human liver, the presence of blood vessel networks suggested that the adoption of essential cells could significantly improve the function of the 3D printed model.

In order to integrate a perfusable vascular network into the liver, in 2018, Prof. Dufva developed a four-arm branched vascular network printing strategy [69]. The hardness and density of the obtained 3D printed liver model are the same as the native liver, and the liver model's vascular networks could simulate the native liver's physiological environment for a long time. The most important innovation of this printing technology is that the obtained tissue models possess larger blood vessels and suitable cell density, which shows great advantages in manufacturing large tissues and organs.

3D printing using hepatocytes as bioink could be able to better simulate the microenvironment in vivo. Prof. Mao, from Tsinghua University, selected bioink containing HepaRG to print liver tissue and transplant it into mice with liver failure. The results showed that the printed tissue could alleviate liver failure and prolong the survival time of mice [70]. This contribution proved the possibility of 3D printed liver tissue in partially replacing liver function, and also showed its potential in regenerative medicine. In 2022, Cuvellier selected the mixture of primary human hepatocytes and methacrylated gelatin as bioink to print liver tissue. The results suggested that the primary human hepatocytes could be well differentiated over time, and the printed tissue showed vascularized capability after implantation to rats [71]. Recently, Prof. Wang et al. developed a novel biomaterial which could fulfill the mechanical properties of 3D printing and the microenvironment required for living cells at the same time [72]. In the contribution, an ionic liquid was selected as a solvent in collagen's modification processes, which provided quantified norbornene groups for crosslink reactions and 3D printing. The obtained functional collagens show excellent solubility and rheological abilities, which could not only provide microstructure support but also improve cell adhesion, proliferation, and differentiation abilities. By using these novel biomaterials, they successfully fabricated centimeter-scale liver tissues with hyperbranched blood vessel networks inside.

As we all know, stem cells and some adult cells could differentiate into hepatocytes. What's more, compared to well-differentiated liver cells, stem cells and adult cells could be taken from patients themselves, which has a great advantage in immunocompatibility [73]. Prof. Chen, coming from the University of California, prepared a 3D hydrogel-based liver model by embedding PSCs along with HUVECs and stem cells into a micro-scale hexagonal structure [74]. This model cannot only enhance the function of HPCs, but also improve the morphological structure. In addition, it can also improve the expression ability of some specific genes in the liver, thereby increasing the secretion of metabolites. Moreover, the experimental results demonstrated that the obtained 3D printed vascularized liver model has good application performance in the fields of new drugs screening and disease models construction.

At present, it remains a challenge to reproduce the inherent heterogeneity and complex microstructures with a limited print volume to fabricate implantable 3D printed liver tissues. Shin et al. recently proposed a combined multi-material and embedded printing method for fabricating liver tissue that is able to implant and maintain three-dimensional structure in the body (Fig. 3) [75]. They designed a computer-directed pressure valve on the nozzle, enabling layer fluidization up to seven kinds of bioink and could quickly switch among the bioinks (Fig. 3 (A)). The novel technology is able to solve alignment problems when switching to different nozzles, and fabricate various vascular networks with complex shapes. By using the printing strategy, they constructed a vascularized liver tissue, exhibiting albumin secretion function.

Although the vascularization characteristics of the printed tissues are becoming more and more like those of real liver tissue, they are still far from completely simulating real liver [76,77]. Regarding liver regeneration, current 3D printing technology has not been able to maintain the function of human primary hepatocytes over the long-term during culture. At least three problems need to be addressed. Firstly, the culture technology of human primary hepatocytes needs improvement. Secondly, researchers should construct more functional structures such as blood vessels and bile ducts. Thirdly, we should optimize the composition and distribution of liver parenchymal cells and interstitial cells in 3D printed tissues. If we make breakthroughs in all of the above three aspects, the 3D printed vascularized liver tissue will very likely improve the existing bioartificial liver, and even combine with artificial blood vessels and other technologies to achieve organ transplantation. In terms of disease model construction, the fidelity of the current vascularized liver tissue response to drugs has not been supported by sufficient data and it is not yet able to simulate the immune microenvironment. Further studies are needed in the future to improve consistency with actual tissue response to drugs, including the regulation of various factors in the microenvironment, involvement of immune cells, and simulation of systemic reactions. If clinical translation could be achieved, drug efficacy and side effect testing through 3D printed liver models will play an important reference role in clinical decision-making.

2.3. Kidney

The kidney is an important organ to maintain the balance of the human's internal environment. It excretes metabolites, regulates blood volume and blood pressure, maintains electrolyte and acid-base balance, and secretes a variety of active substances to maintain normal physiological functions. The blood supply of the kidney is very rich, and in the

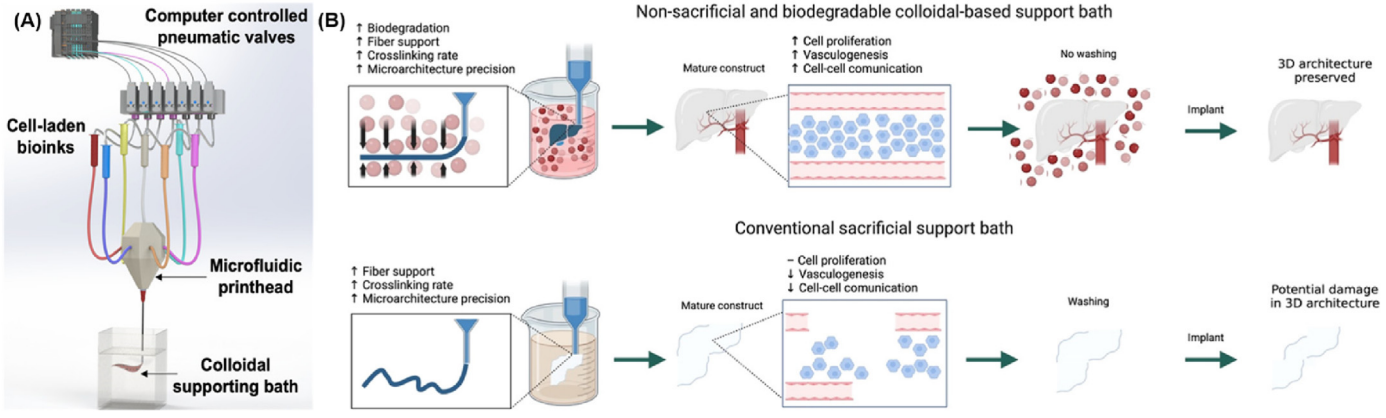


Fig. 3. (A) Nozzle design of a computer-directed pressure valve. (B) The contrast effects of non-sacrificial biodegradable colloidal-based support bath and conventional sacrificial support bath in 3D printing [75]. Reproduced with permission. Copyright 2022, American Chemical Society.

resting state, kidney blood flow accounts for about 20 % of the cardiac output. Such high blood flow requirements of the kidneys are closely related to the formation of urine. Consistent with this function, the renal circulatory system has a unique two-stage capillary blood tube network with a glomerular capillary network in addition to the same capillary network involved in tissue exchange as normal blood vessels. The special structure of the glomerular capillary makes it have high pressure, high perfusion, and high permeability, which are the basis for the completion of ultrafiltration. The glomerular capillary plexus is a network of arterial capillaries curved into balls. After entering the renal capsule, it divides into 4–5 branches and continues to divide into the loops of capillary lobules. The capillaries of each lobule converge to form an outflow arteriole and leave the glomerulus. The entering arterioles are shorter and thicker than the outgoing arterioles, thus maintaining a higher blood pressure in the glomeruli.

The most important function of the kidney is to filter toxins from the blood; thus, it possesses well-developed vascular network [78]. The fabrication of a bionic kidney is a huge engineering feat, requiring the integration of vascular networks, specific cells, microtubule structures, and biomaterials [79–82]. Due to the complex microstructure of kidney tissue, there were not many breakthroughs about biomimetic kidneys in the early years [83]. Recently, 3D printing technology, famous for building fine structures, provides opportunities to manufacture functional kidney tissues [84–86].

The initial application of 3D printing technology in kidney tissue was the manufacture of proximal renal tubules. The tubule structure printed by Moisan and Lewis et al. was completely embedded in the extracellular matrix, and finally a perfusable proximal renal tubule structure was formed on the microfluidic chip [87]. However, this strategy has some limitations in the application of in vitro models due to their ink. In 2020, Kim and Cho et al. mixed the acellular renal extracellular matrix with sodium alginate to produce a new bio-ink and used 3D coaxial printing technology to manufacture perfusable proximal renal tubules and vascular structures (Fig. 4) [88]. As a result, HUVECs and proximal renal tubular epithelial cells were express successfully.

Extrusion process-based 3D printing technology could significantly increase nephron production per starting cell by regulating precise biophysical properties (organoid size, cell number, and microstructural conformation), resulting in a uniformly differentiated functional kidney tissue structure. By using this technology, Prof. Little et al. fabricated a kidney with the advantages of fast and high throughput [89,90]. The above 3D printed kidney possesses the basic units of human kidney structure and function. Moreover, it was utilized to evaluate the relative toxicity of aminoglycosides, verifying the superiority of this 3D printed organoid for drug detection.

3D printing technology completes the design of specific structures and enhances direct contact between cells so that the structural role of interstitial cells can be played. However, most organoids are cultured in a static environment, and there are problems with limited blood vessels and immature cell development. The 3D printed microfluidic chip is able to form an environment for flow culture of organoids in vitro, and could finally construct a perfusable vascular network of kidney organoids [91,92]. In 2019, Prof. Lewis and colleagues created a more optimized 3D proximal tubule model with well-developed vascular networks [93]. It should be noted that they employed a closed-cycle perfusion system to investigate the function of renal reabsorption.

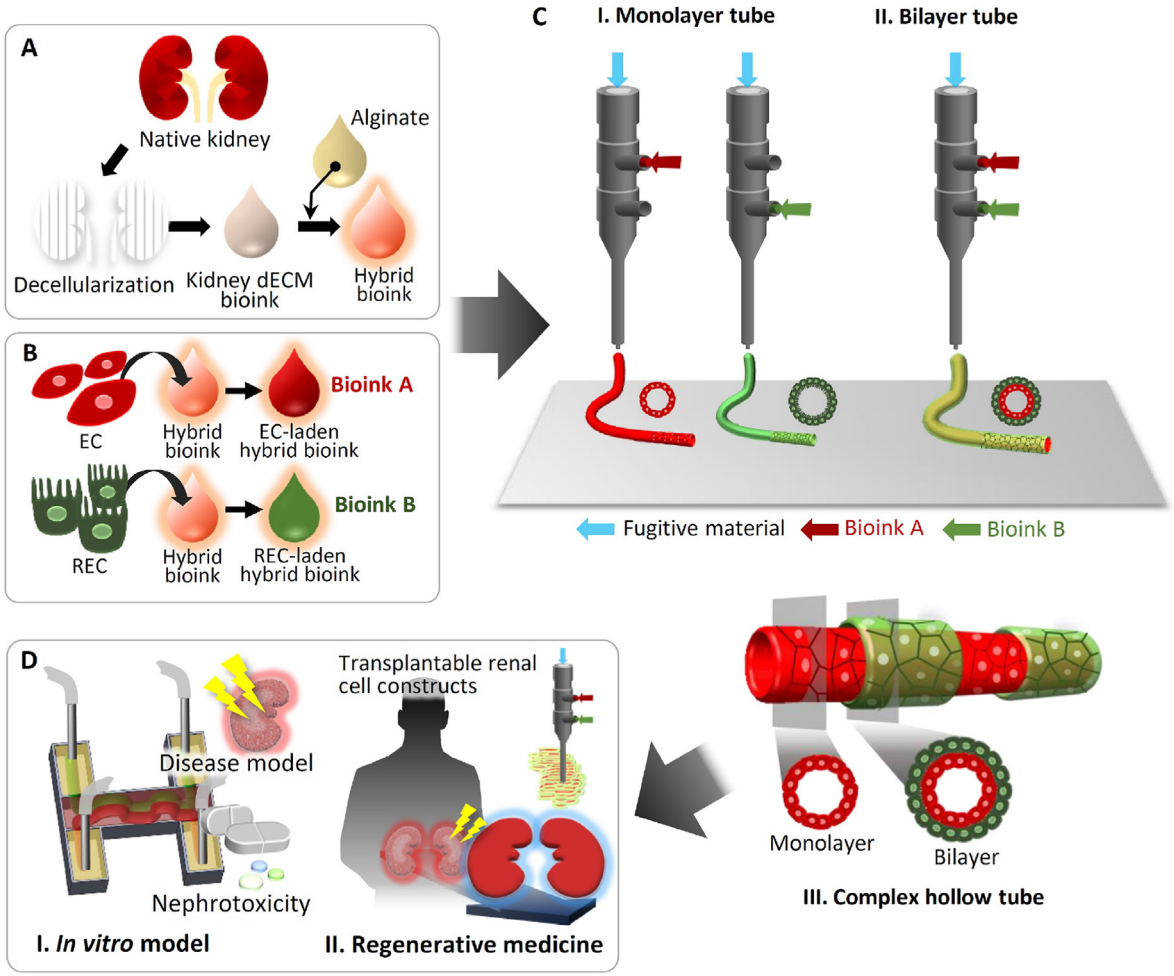


Fig. 4. Fabrication of kidney tissue with well-developed vascular networks by using 3D printing technology [88]. Copyright 2020, Elsevier.

As known to all of us, the cells used in biological ink should be able to realize functions of the targeted cells and expand to large-scale tissue printing quantity. Moreover, the extracellular matrix is able to supply mechanical support and a biomimetic microenvironment for cells' survival and differentiation. Based on the above guidelines, Prof. Kim et al. developed an in vitro strategy for culturing human PSCs derived renal organoid using renal decellularized extracellular matrix hydrogel, which affords tissues with extensive vascular networks and their own ECs [94]. In the same year, Prof. Lewis et al. reported the fabrication of vascularized 3D printed tissues by inducing stem cell differentiation [95]. They demonstrated that the patterned printed tissues could be fabricated by inducing overexpression of transcription and differentiating human-induced PSCs into various types of cells simultaneously. As a result, the vascularized organoid was fabricated by using this orthogonal induction differentiation. Therefore, these methods have great potential to expand 3D printing for building human kidney organ.

Although 3D printed kidney tissue has made rapid development in the past few years and has shown certain application performance for kidney disease, such as drug screening, it is still a long way from clinical organ transplantation [96–99]. The formation of a vascularization network is critical to the survival of cells and further affects the restoration of the structure and function of 3D printed renal tissues and organs; however, the blood supply for large grafts remains a challenge. We suggested that 3D printing has great potential to fabricate vascular microchannel networks of mutual communication through computer modeling, and integrate the above research methods to build a mature and well-perfused vascular network in 3D printed kidney organ [100].

2.4. Lung

The lungs have two sets of blood circulation systems, one is pulmonary blood circulation, and the other is body blood circulation. The pulmonary artery is an important blood vessel for pulmonary circulation. It starts in the right ventricle, travels with the bronchi into the lungs, and branches repeatedly as the bronchi branch, forming a network of capillaries around the alveoli. In general, the right ventricle receives venous blood flowing back to the heart from the superior and inferior vena cava, enters the right ventricle through the heart valve, and then enters the pulmonary artery through the heartbeat. The pulmonary artery differs from other arteries in that it carries venous blood. Another blood vessel of pulmonary circulation is the pulmonary vein. When the venous blood in the pulmonary artery exchanges oxygen and carbon dioxide in the lung, the arterial blood will flow back to the left atrium and left ventricle of the heart through the pulmonary vein and be transported to the capillaries of the whole body. The systemic circulation of the lungs requires bronchial arteries and bronchial veins. The bronchial arteries are relatively small and emanate from the beginning of the thoracic aorta. As the bronchus enters the lung, nutrients are delivered to the bronchial wall, pulmonary vascular wall, visceral pleura, and other parts through blood circulation [101,102].

Due to the great ability of 3D printing technology to fabricate vascularized tissues, biomimetic lung engineering has made some progress in recent years [103–106]. In 2019, Prof. Stevens and Prof. Miller proposed a novel 3D printing strategy that adopts a projector emitting light from below to display successive 2D slices of the model [107]. These printed 3D structures are soft and biocompatible, and have a fine internal structure (resolution: 10–50 μm). In addition, they used a food-grade dye additive as a light absorber, limiting the curing of the hydrogel material to very fine layers. In this way, soft hydrogel structures with complex internal vascular structures could be fabricated in a matter of minutes, and the vascular structures in the obtained alveolar model are stiff enough to prevent bursting when perfusing blood. What's more, the model is able to withstand the simulation process of inhalation and exhalation. In addition, when the red blood cells flow through "blood vessel' fabricated by this strategy, they can effectively obtain oxygen from the "lungs' of breathing, which is close to the function of the human alveoli (Fig. 5 (A)). The vascularized tissues printed by the above method are able to survive in mice, which proves that blood vessels could effectively transport nutrients to tissue cells. Therefore, the afore-mentioned 3D printing technique is a tremendous accomplishment in the realm of vascularized lung tissue manufacturing [46].

In 2021, Prof. Jung and Lee et al. optimized inkjet printing process for the improvement of resolution and accuracy of 3D alveolar model [108]. As exhibited in Fig. 5 (B), the advanced technique enables precise layer-by-layer regulation of spatial arrangement and population of various kinds of cells, rebuilding the complex microvascular network and microstructure as well as morphology of the alveoli. By using the strategy, the team printed ultra-thin three-alveolar models that exhibited surfactant secretion functions and barrier integrity of the alveoli. In the same year, Prof. Yeong proposed a novel strategy to improve the stability and repeatability of the 3D printing process [109]. They adopted a polyvinylpyrrolidone-based bioink and a nozzle with a diameter of 300 μm to print human three-layer alveolar lung models consisting of various human cells. The living cells could survive and proliferate in the obtained 3D printed models, providing a successful example for the manufacturing of a human alveolar lung model. In 2023, Prof. Rheey and Jang fabricated a sophisticated lung cancer model with well-developed vascular networks using a 3D printing strategy [110]. The model consisted of patient-derived lung cancer organoids and lung fibroblasts; thus, assessing of effectiveness in the model is conducive to confirming the curative effect of drugs for lung cancer patients along with fibrosis.

Based on requirements for research and drug screening, in vitro vascularized lung organ chips have made great progress in recent years. In 2015, Rothen-Rutishauser et al. first reported 3D printing of lung organ-on-a-chip models containing a pneumo-blood tissue barrier, which consisted of epithelial and EC layers [111]. Compared to manual cell inoculation, the 3D printing strategy allows for more accurate cell localization, resulting in a thinner and more uniform cell layer. In 2019, Prof. Jeon and Cho et al. developed a lung organ chip using extrusion printing [112]. They used decellularized extracellular matrix (denoted as dECM), which comes from the adhesive membrane of the trachea, as a bio-ink and confirmed that dECM enhances angiogenesis. In 2023, Jung et al. selected an inkjet printing strategy to fabricate a biomimetic lung chip model with a three-layer tissue and a micrometer-grade thickness [113]. It should be noted that the obtained organ-on-a-chips is a multi-purpose platform, capable of being applied to a variety of organ models.

At present, 3D printing technology plays a very important role in the manufacture of vascularized lung tissues, realizing the construction of elaborate 3D heterogenous tissues, and promoting the advancement of lung tissue engineering [114]. Compared with traditional lung tissue engineering methods, the lung tissue obtained by 3D printing is able to better simulate the in vivo environment, which could provide a more appropriate solution for clinical drug screening [115–119]. Although the past five years have witnessed some achievements in 3D printing of vascularized lung tissue, there are a lot of problems that should be resolved before the clinical transplantation of printed lung tissues [120]. Firstly, the precision of 3D printing technology should be improved to better mimic the microstructure of natural lung tissue. Secondly, better bioinks are urgent for the improvement of the biocompatibility and breathability of artificial tissues, providing a suitable environment for cell differentiation and growth. Furthermore, lung tissue, which contains both gas exchange networks and blood supply networks, has a more complex structure, so more effective methods for promoting micro-angiogenesis should be developed.

2.5. Penis

The arterial system of the penis is divided into shallow and deep parts. The superficial artery comes mainly from the external artery, which can be further divided into dorsal and ventral branches, responsible for supplying the skin and foreskin of the penis. The deep artery comes from

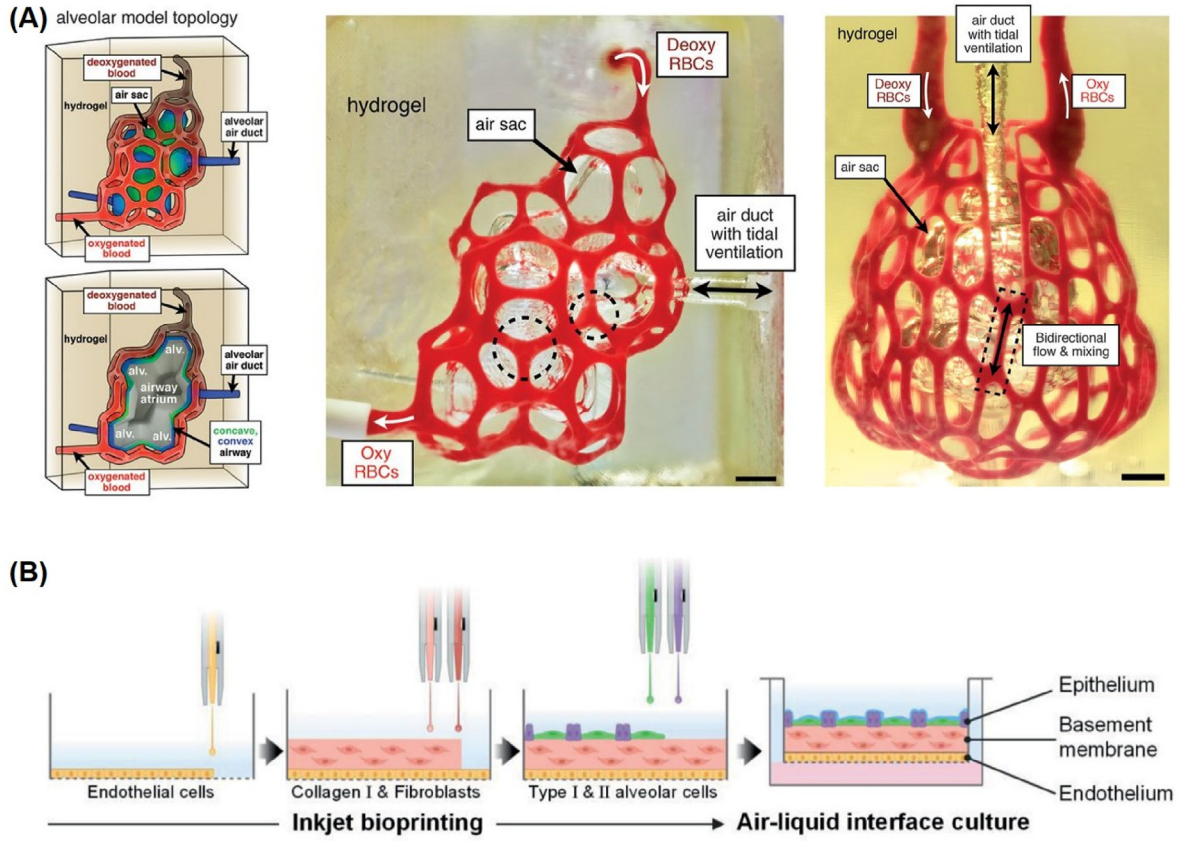


Fig. 5. (A) Vascularized alveolar model obtained by 3D printing [107]. (B) An alveolar barrier model was prepared by 3D printing technology in a layer-by-layer strategy [108]. Reproduced with permission. Copyright 2019 [107], American Association for the Advancement of Science, and adapted from an open-access source [108].

the internal artery and divides into the bulbar and urethral arteries near the bulbar of the urethra, and then further divides into the dorsal penile artery and the deep penile artery. The venous system of the penis is also divided into three levels: superficial, middle, and deep. The superficial veins mainly include the superficial dorsal veins, the middle veins include the deep dorsal veins and circumflex veins, and the deep veins include the cavernous veins and the foot veins. These veins intertwine with the arterial system and form the blood circulation system of the penis. Both the arterial and venous systems play an important role in the blood supply to the penis. The arterial system is responsible for providing nutrients and oxygen that promote the growth and function of the penis, while the venous system is responsible for recycling blood and metabolic waste to keep the penis healthy.

The penis is an important male sexual organ, mainly composed of penis corpus cavernosum and cavernous urethra, with base fascia and skin [121]. Lesions and damage to the cavernosum tissue would cause erectile dysfunction, resulting in decreased fertility and unsatisfactory sexual life. The erection could be simply summarized as the congestive process of the penis corpus cavernosum, and it can also be considered as the result of the dilation of the penis artery and the relaxation of the smooth muscle of the corpus cavernosum. Therefore, the blood vessels play a vital role in penis tissue.

Although 3D printing has received widespread attention in the field of artificial tissues, the fabrication of penile tissues using 3D printing strategies is still in its early stages [122,123]. In recent years, our group has been developing biocompatible and histocompatible hydrogels, and exploring their application performance as tissue engineering materials [124]. In 2020, our group prepared a novel 3D printing ink which consisted of methacrylamide-modified gelatin and methacrylated hyaluronic acid along with stem cells to improve biocompatibility and angiogenesis ability (Fig. 6) [125]. In the work, a 3D print-ultraviolet photo cross-linking strategy was used to fabricate the hydrogel scaffold, and heparin was deposited onto the scaffold to stimulate the formation of neovascularization at the implantation site. The experiment results suggested that the obtained hydrogel scaffold is in favor of vascular generation, and obvious new angiogenesis could be observed after implantation in mice. Moreover, after the transplantation of hydrogel scaffolds to injured corpus cavernosum, rabbits could regain erectile function and even restore fertility. This work not only successfully first repaired penis corpus cavernosum defects using stem cells and 3D printing technology but also took an important step towards the development of vascularized bioengineering scaffolds. It should be noted that, in addition to being applied to the cavernous tissue of the penis, this 3D printed tissue also has the potential to repair other tissues containing blood vessels. Recently, we further constructed an artificial tunica albuginea by using strain-strengthened hydrogels composed of aligned and curled fibers, which showed the ability to repair damaged penile tissue in a pig model [126].

3D printing penile tissue is still in its infancy; however, the development of the technology has important implications for treating erectile dysfunction. In the future, we will continue to explore novel bioinks that could print vascularized penile tissue and optimize the printing process, with a view to fabricate penile tissues and organs that have similar functions to human penises and can even be used for clinical transplantation.

3. Conclusion and future perspective

3D printing technology is playing an increasingly important role in manufacturing artificial tissues and organs. At present, researchers have

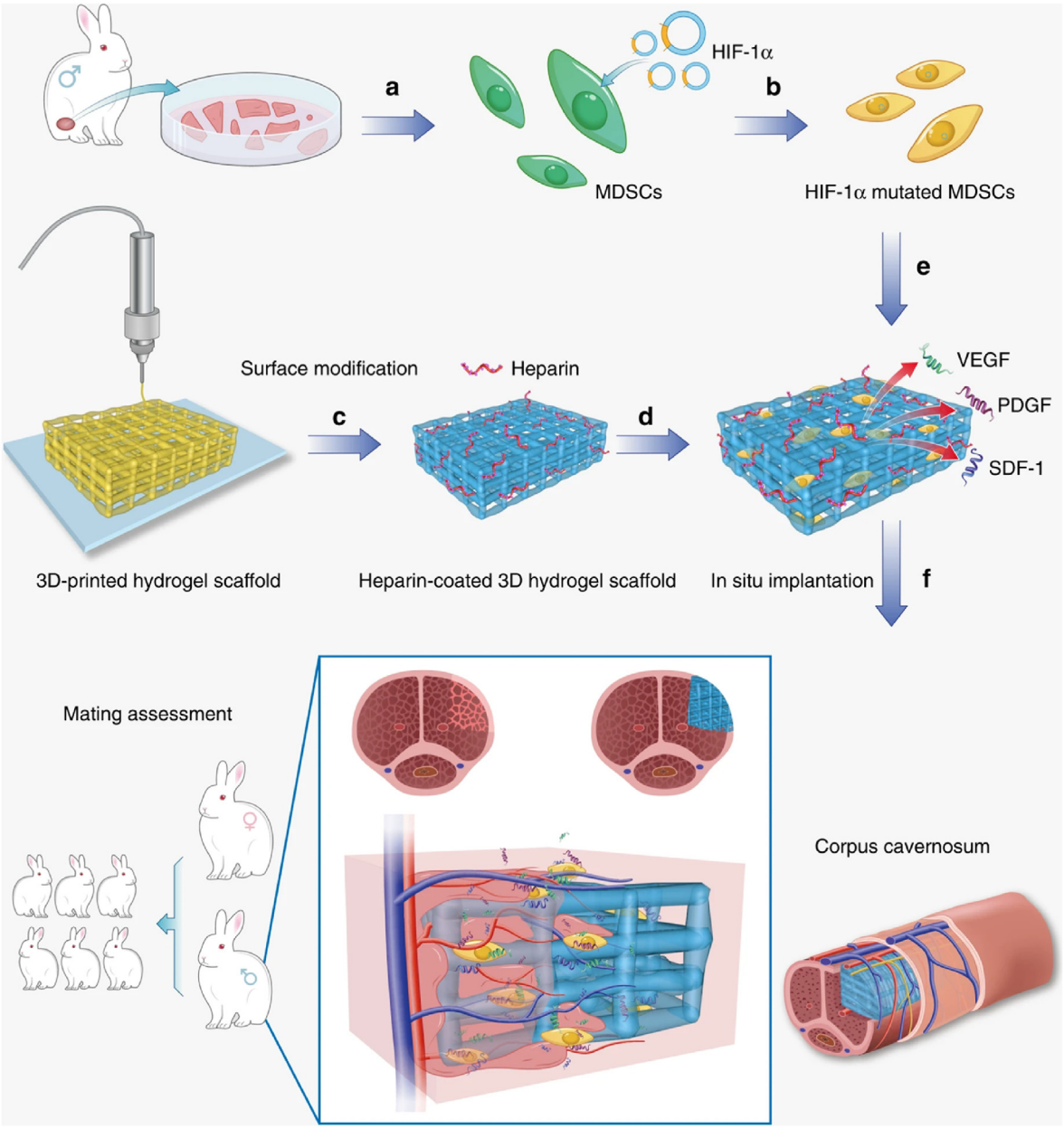


Fig. 6. Repairing rabbits' corpora cavernosa penis by in situ implantation of 3D-printed hydrogel scaffold [125]. Adapted from an open-access source.

successfully constructed biomimetic vascularized heart, liver, lung, kidney, and corpus cavernosum of the penis, *etc*. However, most of them are thin tissues or functional units. This is mainly because the cell number required to construct a complete organ and the vascular network required to maintain the activity of large tissue is difficult to meet the requirements. With the continuous progress of this technology, in recent years, there have been reports that have tried to print bionic organs, showing the ability of the technology in tissue and organ manufacturing, laying a technical foundation for the future realization of large-size tissue fabrication that could be transplanted.

For 3D printing, the bioinks, printing methods, and printing equipment would affect the structure and performance of the obtained tissues. Currently, the main challenges include the following aspects. Firstly, it is urgent to develop tissue-specific bioinks that could not only guarantee the mechanical properties of printed tissues, but also provide the microenvironment in favor of cell survival, which is beneficial to improving cell survival and adapting to the development and maturation of different tissues. Secondly, advanced printing strategies need to be developed to achieve high-precision printing (such as single-cell level), simultaneous multi-material printing, and fast printing. Thirdly, for the obtaining of tissues and organs with biological functions, it is necessary to effectively achieve vascularization network construction, nerve innervation, and immune regulation of printed tissues and organs. Moreover, we also need to make legal preparations, and regulatory agencies should establish and improve the standards and norms for the evaluation of the safety, efficacy and risks of printed tissues and organs.

In conclusion, although the clinical applications of 3D printed vascularized tissues and organs are still in their infancy, the current achievements give rise to confidence in the future of 3D printing, promising to produce functional tissues and organs with biomimetic vascular networks. We believe that with continuous progress of printing technologies and perfection of bioinks, 3D printing will bring revolutionary changes to tissue engineering.

CRediT author statement

Ke Zheng: Literature collection, Writing-original draft; **Muyuan Chai:** Writing-review, Proofreading; **Bingping Luo:** Literature collection, Proofreading; **Kezhao Cheng:** Literature collection; **Zhenxing Wang:** Literature collection; **Nan Li:** Writing-review, Supervision; **Xuetao Shi:** Conceptualization, Writing-review & Editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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